

Braillix - Master Bill of Materials

Supersedes docs/SHOPPING_LIST.md (that file still lists 4mm pen springs and 2mm bearing balls - both wrong since v7.1). Buy from this file only. Scope: **1 Brain Pod + 1 Cell**, enough to make one dot go up and come back down. Budget ₹15,000 - already spent ₹935 (J.B. Enterprises). Fastener quantities are **derived from the .scad sources and verified**, not estimated - each row names the feature it serves. Prices are realistic Indian estimates (Amazon.in / robu.in / local market), marked ~ where not quoted from a specific listing.

Status legend

	Meaning
[OK] HAVE IT	Already owned - from the ₹935 bill or stated in hand
[X] NEED TO BUY	Not owned. Where + estimated cost given
[!] HAVE BUT WRONG	Owned, but the wrong spec or not enough

1. Electronics

Component	Status	Action Needed
ESP32 DevKit V1 (30-pin, USB-C)	[OK] HAVE IT	-
5V/3A power adapter	[OK] HAVE IT	-
Inline female DC pigtail jack	[OK] HAVE / CURRENT POD CHOICE	Connector body remains outside; only the red/black leads pass through the lid slot and are strain-relieved on the internal tie post. Polarity measured correct 2026-08-01, red = positive. Prove slot size with <code>pigtail_slot_coupon.gcode</code> .
Threaded 5.5x2.1mm female panel-mount jack	? NOT REQUIRED FOR THIS REVISION	The old guessed panel-jack cradle was removed. Reconsider only in a future enclosure redesign.
28BYJ-48 stepper motor (5V)	[OK] HAVE IT / MEASURED	Fit-critical caliper readings recorded, including M5=3.0mm. Vertical stack still needs one coordinated CAD pass.
ULN2003 driver module	[OK] HAVE IT	Nominally fits the 14mm pocket only with wires soldered flat (~12mm). Vertical Dupont headers will not fit; final board/tail measurement and retention are still required.
Hall sensor - MH-Sensor-Series (blue module)	[!] HAVE BUT WRONG FORM FACTOR	The whole blue PCB will not fit. v7.5 provides a 4.5 x 3.5 x 1.6mm underside pocket for its bare TO-92 sensor. M11b is exactly 1.6mm, so dry-fit before gluing. Desolder the sensor and run three wires back; use analog AO behavior for homing.
Neodymium magnets 8x1mm (docking)	[OK] HAVE IT	10 purchased. 4 needed per cell (2 per +/-X face).
Homing magnet 3x1mm (for the cam)	[X] NEED TO BUY	1 per cell. The blind pocket is 1.2mm deep and leaves 0.8mm of cam floor. An 8x1mm docking magnet is too wide.
Tactile switches 6x6x5mm	[OK] HAVE IT	3 needed (pod nav buttons). Not required for a Tier-0 demo.
Female header strips 1x15	[X] NEED TO BUY	2 for the pod DevKit mount. ~₹40. Not needed for a breadboard demo.
USB-C cable (data, not charge-only)	[X] NEED TO BUY / verify	Must carry data . Many charge-only cables look identical and silently fail to flash. ~₹150.
Resistors / capacitors	[OK] NOT NEEDED	4.7k I2C pull-ups are only for the multi-cell I2C bus, which is not in the demo. No protection parts needed for a single cell.

Component	Status	Action Needed
Pogo pin connectors	?? DEFERRED	Do not buy yet. Require a measured flush carrier and $\geq 1A$ /contact continuous (prefer 2-3A for power). The four-wire architecture also needs a per-cell 3.3V expander interface.

2. Mechanical & fasteners

Counts derived from the SCAD sources and verified against them.

Component	Qty	Status	Serves / Action
M2.5 x 25mm bolts	4	[X] NEED TO BUY	top_plate counterbores -> base_plate standoffs -> outer_box corner bosses. Verified: 25mm gives ~9.5mm engagement. ~?60
M4 x 5mm thread-forming screws	2	?? SELECT AFTER COUPON	Motor ears into the dia 3.3/3.4/3.5 coupon-selected through pilots. Do not use M4x6 or M4x10 until the real ear and screw point prove $\geq 0.5mm$ cam clearance. No nuts.
M2 x 8mm self-tap	2	[X] NEED TO BUY	Pod lid -> shell bosses. [!!] NOT M2x6 - through a 4mm lid that leaves only 2mm of thread; 8mm gives 4mm. Verified. ~?40
M2 x 6mm self-tap	4	?? DEFERRED	Muscle-board bosses. Not used in the prototype (ULN2003 sits on the pocket floor).
2mm OD micro compression springs	6 + spares	[X] NEED TO BUY	[!!] CRITICAL, no substitute. 2.0mm OD, ~0.3mm stainless wire, ~4mm free length. Pen springs (4mm) cannot work - braille rows are 2.6mm apart. Buy an assortment kit (200-400pcs) so you're not betting on one guess. ~?500

3. Consumables

Component	Status	Action Needed
Solder wire	[X] NEED TO BUY	0.8mm, 60/40 rosin-core. Not plumbing solder. ~?150
Flux paste	[X] NEED TO BUY	Not strictly required, but makes soldering to ULN2003 pins far easier for a beginner. ~?100
Superglue (CA) or 5-min epoxy	[X] NEED TO BUY	Glues the resin dot_insert into the PETG top plate, and the magnets. Epoxy recommended - it gives you time to seat the tile square. ~?80
Heat-shrink tubing (1-3mm assorted)	[X] NEED TO BUY	Insulating soldered joints. ~?80
Electrical tape	[X] NEED TO BUY	~?30
Silicone grease	[X] NEED TO BUY	Cam tracks, to reduce stall risk. A tiny amount. ~?80
Dupont jumper wires	[OK] HAVE IT	2m wire + spares purchased. For counts see below.
Breadboard	[X] NEED TO BUY	Yes, still needed - the Tier-0 demo is breadboard-based, and it is the safest way to bring up wiring before anything is soldered. Half+ size. ~?100

Dupont counts for the demo wiring (13 connections total): **M-F x8** (ESP32 header -> breadboard/module), **M-M x5** (rail to rail, module to rail). F-F not needed. If your 2m wire is bare, you need crimps or you solder directly.

4. Tools

Tool	Status	Action Needed
Soldering station	[X] NEED TO BUY	[!!] Not owned (an earlier revision of this file wrongly said it was). Buy a 60W temperature-controlled "936" station with a 2.4mm chisel tip , ~?1,300 - NOT a plain ?200 pencil iron. Full reasoning in ELECTRONICS_PLAN.md Part 9.
Digital multimeter	[OK] HAVE IT (bought 2026-08-01)	Jack polarity measured and confirmed correct - red = positive. The largest single risk to the ESP32 is retired.
Digital calipers	[OK] IN HAND / USED	Motor, Hall thickness, and ESP32 row-pitch readings recorded on 2026-07-31.
Wire stripper / cutter	[X] NEED TO BUY	~?200
Desoldering pump or wick	[X] OPTIONAL	Useful if you need to lift the TO-92 off the hall module, or fix a bridge. ~?120
Tweezers	[X] OPTIONAL	Strongly recommended for 2mm springs and 1mm linkages. ~?80
Small drill bits (1.5/1.7/2.5mm)	[X] OPTIONAL	FDM holes print 0.2-0.3mm undersize; these clean them out. ~?150

Soldering difficulty - the honest read

Nothing in the demo build requires fine-pitch work. The hardest joint is soldering wires flat onto the ULN2003's 2.54mm header pads - comfortably within basic skill.

The **only** fine-pitch work in the whole project is the custom muscle board (TQFP-32 at 0.8mm pitch, SOIC-16, 0402 passives). That is **beyond basic soldering** - order it assembled (PCBA) or stay on the ULN2003 module. It is not needed for the demo either way.

5. What to actually spend

Path	Contents	Cost
Cheapest working demo	multimeter, breadboard, solder wire, strippers, USB-C cable	~?1,100
+ mechanism	above + springs, homing magnet, fasteners, glue, calipers	~?2,900
Everything in this BOM	all of the above + optional tools + consumables	~?3,900

Project total if you buy everything: ?935 + ~?3,900 = ~?4,835 of ?15,000. Comfortable.

Deliberately excluded: the resin print order. Do the cheap PETG proving print and make one dot move *before* spending on resin. Every downstream decision rests on a mechanism that has never been physically tested.

6. Buy these first (this week)

- 1 **Multimeter** (~?500) - unblocks safe power-up. Nothing else can happen safely without it.
- 2 **Breadboard + USB-C data cable + solder wire** (~?400) - everything needed for the Tier-0 demo.

No panel-mount jack is required for the current revision; use the owned pigtail and prove its cable-only slot first.

Calipers are now in hand and the motor, Hall thickness, and ESP32 row-pitch measurements are recorded.

7. Unverified - measure before buying or printing

#	What	Feeds
1	Motor vertical stack re-derivation from the completed owned-part readings	base_plate, mid_plate, cam bore
2	Smallest pigtail cable slot that passes both insulated leads without damage	power_cable_slot_w/h in the pod lid; use the 5x3 / 6x4 / 7x5 coupon
3	Hall module PCB dimensions (or switch to bare TO-92)	base_plate hall pocket
4	Actual magnet diameter/thickness from the bill	dock magnet pockets (CAD assumes 8x1mm)
5	ESP32 board length/width if the catalogue envelope proves tight; pin-row pitch is measured 25.6mm	pod board envelope
6	Pogo connector, once chosen	outer_box pogo carrier pocket